Часть 1. Коммутатор и 2 компьютера

Для настройки используем сеть 192.168.1.0/24

Настройка ip первого компьютера

```
PC1> ip 192.168.1.10 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0

PC1> save
Saving startup configuration to startup.vpc
.  done
```

Настройка ip второго компьютера

```
PC2> ip 192.168.1.20 255.255.255.0
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0

PC2> save
Saving startup configuration to startup.vpc
.  done
```

Проверяем, что все записалось

```
PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 255.255.255.0
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 25868
RHOST:PORT  : 127.0.0.1:25869
MTU         : 1500
```

Также для второго компьютера

```
PC2> show ip

NAME        : PC2[1]
```

```
IP/MASK     : 192.168.1.20/24
GATEWAY     : 255.255.255.0
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 25870
RHOST:PORT  : 127.0.0.1:25871
MTU         : 1500
```

Пингуем с первого компьютера второй компьютера

```
PC1> ping 192.168.1.20

84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=2.602 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=64 time=0.206 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=64 time=0.331 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=64 time=1.017 ms
^C
PC1>
```

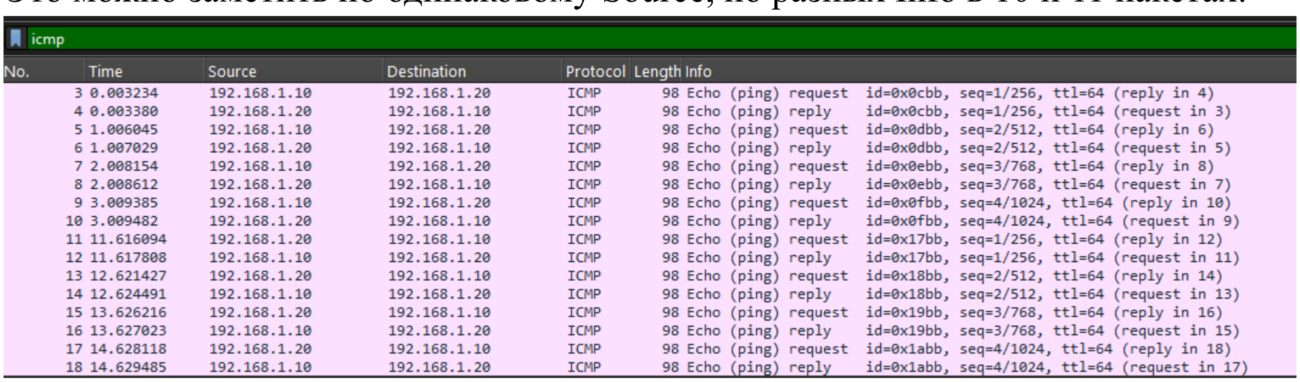
Анализ WireShark

С первого компьютера(192.168.1.10) пингуем второй(192.168.1.20) 4 раза. Потом Второй(192.168.1.20) пингует 4 раза первый(192.168.1.10).

Как можно видеть по файлу захвату первые 4 пары request-reply идут от первого ко второму компьютеру, что говорит столбец Source и Destination, а также Info.

С 3-10 – это пинги от первого ко второму, а с 11 по 18 – это от второго к первому.

Это можно заметить по одинаковому Source, но разных Info в 10 и 11 пакетах.


icmp

No.	Time	Source	Destination	Protocol	Length	Info
3	0.003234	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) request id=0x0cbb, seq=1/256, ttl=64 (reply in 4)
4	0.003380	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0cbb, seq=1/256, ttl=64 (request in 3)
5	1.006045	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) request id=0x0dbb, seq=2/512, ttl=64 (reply in 6)
6	1.007029	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0dbb, seq=2/512, ttl=64 (request in 5)
7	2.008154	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) request id=0x0ebb, seq=3/768, ttl=64 (reply in 8)
8	2.008612	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0ebb, seq=3/768, ttl=64 (request in 7)
9	3.009385	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) request id=0x0fbb, seq=4/1024, ttl=64 (reply in 10)
10	3.009482	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0fbb, seq=4/1024, ttl=64 (request in 9)
11	11.616094	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x17bb, seq=1/256, ttl=64 (reply in 12)
12	11.617808	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) reply id=0x17bb, seq=1/256, ttl=64 (request in 11)
13	12.621427	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x18bb, seq=2/512, ttl=64 (reply in 14)
14	12.624491	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) reply id=0x18bb, seq=2/512, ttl=64 (request in 13)
15	13.626216	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x19bb, seq=3/768, ttl=64 (reply in 16)
16	13.627023	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) reply id=0x19bb, seq=3/768, ttl=64 (request in 15)
17	14.628118	192.168.1.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x1abb, seq=4/1024, ttl=64 (reply in 18)
18	14.629485	192.168.1.10	192.168.1.20	ICMP	98	Echo (ping) reply id=0x1abb, seq=4/1024, ttl=64 (request in 17)

Часть 2. Маршрутизатор и 2 компьютера

Маршрутизатор разделяет сеть на разные широковещательные домены. Используем две подсети:

- Подсеть PC1: 192.168.1.0/24
- Подсеть PC2: 192.168.2.0/24

Настройка ip для PC1

```
PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> save
Saving startup configuration to startup.vpc
.  done
```

Также для PC2

```
PC2> ip 192.168.2.20 255.255.255.0 192.168.2.1
Checking for duplicate address...
PC2 : 192.168.2.20 255.255.255.0 gateway 192.168.2.1

PC2> save
Saving startup configuration to startup.vpc
.  done
```

Настройка маршрутизатора на базе c3640. В нем создаем две разные подсети на два порта.

```
Press RETURN to get started!

-5-COLDSTART: SNMP agent on host R1 is undergoing a cold start
*Mar  1 00:00:09.943: %LINK-5-CHANGED: Interface FastEthernet0/0, changed
state                          to administratively down
*Mar  1 00:00:09.947: %LINK-5-CHANGED: Interface FastEthernet1/0, changed
state                          to administratively down
R1#
R1#enable
R1#configure terminal
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:01:58.415: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed
state t                          o up
*Mar  1 00:01:59.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthern                          et0/0, changed state to up
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:02:39.023: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed
state t                          o up
*Mar  1 00:02:40.023: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthern                          et1/0, changed state to up
R1(config)#end
R1#wri
*Mar  1 00:02:43.295: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
```

Проверяем назначение подсети на порт

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0            192.168.1.1     YES manual up                    up
FastEthernet1/0            192.168.2.1     YES manual up                    up
R1#
```

Пингуем с PC2 на PC1

```
PC2> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=63 time=21.025 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=63 time=11.335 ms
```

```
84 bytes from 192.168.1.10 icmp_seq=3 ttl=63 time=16.897 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=63 time=13.779 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=63 time=13.716 ms
^C
PC2>
```

Анализ WireShark

С второго компьютера(192.168.2.20) пингуем второй(192.168.1.10) 5 раза. Потом первый(192.168.1.10) пингует 5 раза первый(192.168.2.20).

Как можно видеть по файлу захвату первые 4 пары request-reply идут от первого ко второму компьютеру, что говорит столбец Source и Destination, а также Info.

icmp

No.	Time	Source	Destination	Protocol	Length	Info
3	16.122325	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x5bbe, seq=1/256, ttl=63 (reply in 4)
4	16.122416	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) reply id=0x5bbe, seq=1/256, ttl=64 (request in 3)
5	17.138705	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x5cbe, seq=2/512, ttl=63 (reply in 6)
6	17.138843	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) reply id=0x5cbe, seq=2/512, ttl=64 (request in 5)
7	18.162661	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x5dbe, seq=3/768, ttl=63 (reply in 8)
8	18.162760	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) reply id=0x5dbe, seq=3/768, ttl=64 (request in 7)
9	19.175524	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x5ebe, seq=4/1024, ttl=63 (reply in 10)
10	19.176074	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) reply id=0x5ebe, seq=4/1024, ttl=64 (request in 9)
11	20.199481	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) request id=0x5fbe, seq=5/1280, ttl=63 (reply in 12)
12	20.200878	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) reply id=0x5fbe, seq=5/1280, ttl=64 (request in 11)
14	28.200706	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x67be, seq=1/256, ttl=64 (reply in 15)
15	28.221579	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x67be, seq=1/256, ttl=63 (request in 14)
16	29.222087	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x68be, seq=2/512, ttl=64 (reply in 17)
17	29.235044	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x68be, seq=2/512, ttl=63 (request in 16)
18	30.235934	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x69be, seq=3/768, ttl=64 (reply in 19)
19	30.258076	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x69be, seq=3/768, ttl=63 (request in 18)
20	31.259035	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x6abe, seq=4/1024, ttl=64 (reply in 21)
21	31.274172	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x6abe, seq=4/1024, ttl=63 (request in 20)
22	32.275593	192.168.1.10	192.168.2.20	ICMP	98	Echo (ping) request id=0x6bbe, seq=5/1280, ttl=64 (reply in 23)
23	32.297024	192.168.2.20	192.168.1.10	ICMP	98	Echo (ping) reply id=0x6bbe, seq=5/1280, ttl=63 (request in 22)